

High-Performance In-Memory Data Wrangling using data.table

Introduction

This project demonstrates high-performance data processing using the **data.table** package in R. A large sales transaction dataset containing **500,000 records** is generated and processed efficiently. The project focuses on data generation, cleaning, manipulation, aggregation, business analysis, and visualization to extract meaningful business insights.

Objectives

- Generate a large sales transaction dataset.
 - Perform data cleaning and preprocessing.
 - Manipulate data efficiently using **data.table**.
 - Calculate revenue, discount, GST, and net revenue.
 - Perform grouping and aggregation.
 - Generate business reports.
 - Visualize important business insights.
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Software Requirements

- R Programming
 - RStudio
 - data.table Package
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Dataset Description

The dataset contains **500,000 sales records** with the following attributes:

- Transaction ID
- Customer ID
- Customer Name
- Gender
- Age
- Product ID
- Product Name
- Category
- Quantity

- Price
 - Discount
 - GST
 - Payment Method
 - City
 - State
 - Order Date
 - Sales Person
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Methodology

Step 1: Dataset Generation

A synthetic sales dataset containing 500,000 records is generated using random values. Customer information, product details, payment methods, cities, and transaction dates are created to simulate real-world sales data.

Step 2: Data Import

The generated dataset is saved as a CSV file and imported back into R using the **fread()** function for high-speed reading.

Step 3: Data Exploration

The dataset is explored by:

- Checking missing values
 - Identifying duplicate records
 - Finding unique customers, products, and cities
 - Displaying dataset structure and summary
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Step 4: Data Cleaning

The dataset is cleaned by:

- Removing duplicate records
 - Renaming the **UnitPrice** column to **Price**
 - Converting **OrderDate** into Date format
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Step 5: Data Manipulation

Various data.table operations are performed, including:

- Selecting columns
 - Filtering records
 - Sorting data
 - Finding top expensive products
 - Finding highest quantity transactions
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Step 6: Revenue Calculation

New business columns are created:

- Revenue
- Discount Amount
- GST Amount
- Net Revenue

These values help analyze the actual sales revenue after applying discount and GST.

Step 7: Indexing

Indexes are created using **setkey()** on Customer ID and Product ID to improve search performance and enable faster data retrieval.

Step 8: Grouping and Aggregation

Business summaries are generated for:

- Customers
- Products
- Cities
- Categories

This helps understand revenue distribution and sales performance.

Step 9: Business Reports

The summarized results are exported as CSV files:

- customer_summary.csv

- product_summary.csv
- city_summary.csv
- category_summary.csv

These reports are stored inside the **Output** folder.

Step 10: Business Analysis

The project identifies:

- Highest revenue city
 - Maximum selling product
 - Top 20 customers
 - Most used payment method
 - Average transaction value
 - Highest revenue category
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Step 11: Data Visualization

The project generates the following charts:

- Transactions by City (Bar Chart)
- Revenue by Category (Bar Chart)
- Top Customers (Bar Chart)
- Payment Method Distribution (Pie Chart)

These visualizations provide a clear understanding of sales trends and customer behavior.

Results

The project successfully generated and processed a large sales dataset using the **data.table** package. Business reports and visualizations were created efficiently, demonstrating the advantages of in-memory data processing for handling large datasets.